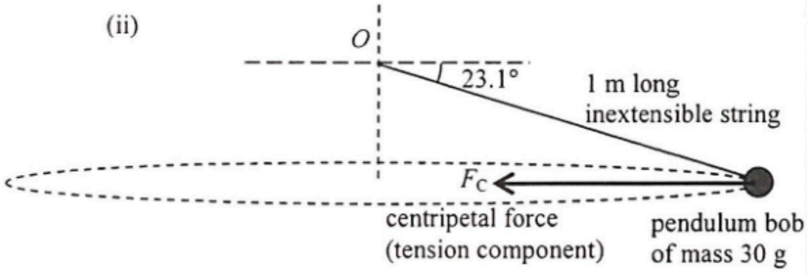
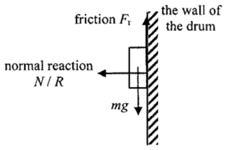
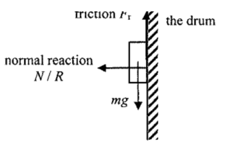
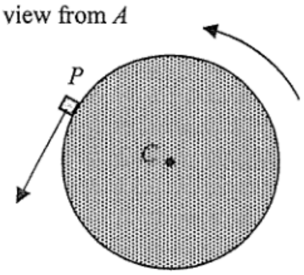
3.	(a)	(i)	Friction f between the tyres and the road.	1A	
			$f = \dfrac{mv^2}{r}$	1M	
			$8000 = \dfrac{1200\ v^2}{45\ \text{m}}$		
			$v = 17.3\ \text{m s}^{-1}$	1A	3
		(ii)	Smaller	1A	
			For the same f, $v^2 \propto r$; when r decreases v decreases.	1A	2
	(b)		(Max) friction / coefficient of friction reduced,	1A	
			not enough to provide the centripetal force / acceleration required for circular motion.		
			Or Tracking or allowed speed lowered.	1A	2

4. (a) (i) Rotation rate $= \dfrac{\omega}{2\pi} = \dfrac{5.0}{2\pi}$ $= 0.795775\ (\text{rev s}^{-1}) \approx 0.80\ (\text{rev s}^{-1})$	1M/1A	Accept: $(0.79 \sim 0.80)$ rev s^{-1}
	1	
(ii) 		
F_C correctly indicated.	1A	
$F_C = mr\omega^2$		
$= (0.03)(1 \times \cos 23.1°)(5.0)^2$	1M	
$= 0.689866\ \text{N} \approx 0.690\ \text{N}$	1A	
($F_C = 0.7033402\ \text{N} \approx 0.703\ \text{N}$ for $g = 10\ \text{m s}^{-2}$)		
<u>OR</u> $T\cos\theta = F_C$ and $T\sin\theta = mg$	1M	
$F_C = \dfrac{mg}{\sin\theta}\cos\theta = 0.689866\ \text{N} \approx 0.690\ \text{N}$	1A	
	3	
(iii) Horizontal component of tension provides the centripetal force, thus tension is larger than the centripetal force.	1M 1A	
<u>OR</u> $T\cos\theta = F_C$ $\Rightarrow T > F_C$	1M 1A	
<u>OR</u> $T\sin\theta = mg$ $\Rightarrow T\ (0.750\ \text{N}) > F_C\ (0.690\ \text{N})$	1M 1A	
	2	
(b) (i) The gravitational force is always perpendicular to the moon's motion / displacement / velocity,	1A	
thus no work is done on the moon by this force (k.e. unchanged)	1A	
	2	
(ii) (The claim is incorrect) as, by Newton's third law of motion,	1A	Accept: action and reaction pair
gravitational force of the same magnitude (but in opposite direction) is acting on the Earth by the moon.	1A	
	2	

		Answer	Marks	Total
4. (a)	(i)		2A	2
	(ii)	Normal reaction (force acting on drum/wall of the drum by the object)	1A	1
(b)	(i)	$\omega = 78.5$ rad s^{-1} rotation speed $= \frac{78.5}{2\pi} \times 60$ $= 749.619782 \approx 750$ (rev min^{-1})	 1M 1A	
		Or $T = \dfrac{2\pi}{\omega} = 0.0800441$ s and $f = \dfrac{1}{T} = 12.493663$ rev s^{-1} $= 749.619782 \approx 750$ (rev min^{-1})	 1M 1A	2
	(ii)	centripetal force $= m\omega^2 r$ $= (2 \times 10^{-5})(78.5)^2(0.25)$ $= 0.03081125$ N ≈ 0.0308 N	 1M 1A	2
	(iii)	$v = r\omega$ $= 0.25 \times 78.5$ $= 19.625 \approx 19.6$ m s^{-1}	 1M 1A	2
	(iv)	I	1A	1

		Answer	Marks	Total
			2A	2
	(ii)	Normal reaction (force acting on drum/wall of the drum by the object)	1A	1
(b)	(i)	$\omega = 78.5$ rad s^{-1} rotation speed $= \frac{78.5}{2\pi} \times 60$ $= 749.619782 \approx 750$ (rev min^{-1})	 1M 1A	
		Or $T = \dfrac{2\pi}{\omega} = 0.0800441$ s and $f = \dfrac{1}{T} = 12.493663$ rev s^{-1} $= 749.619782 \approx 750$ (rev min^{-1})	 1M 1A	2
	(ii)	centripetal force $= m\omega^2 r$ $= (2 \times 10^{-5})(78.5)^2(0.25)$ $= 0.03081125$ N ≈ 0.0308 N	 1M 1A	2
	(iii)	$v = r\omega$ $= 0.25 \times 78.5$ $= 19.625 \approx 19.6$ m s^{-1}	 1M 1A	2
	(iv)	I	1A	1

4.	(a)	(i)	$F = m\omega^2 r = 0.020 \times 6^2 \times 0.50$ $= 0.36$ N	1M 1A 2
		(ii)	Top view from A 	1A 1
	(b)	(i)	same angular speed	1A 1
		(ii)	different / smaller magnitude of centripetal acceleration ($a_Q < a_P$) as Q takes a different / smaller radius ($r_Q < r_P$)	1A 1A 2